

NIKHIL MAHADIK

Mumbai Maharashtra

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Objective

Backend & AI Software Engineer with 4+ years of experience building scalable **FastAPI services**, **multi-agent systems**, and **production-grade RAG pipelines**. Proficient in **REST APIs**, **database design (SQL)**, and **LLM orchestration**. Strong expertise in **asynchronous programming**, **Docker**, **cloud integrations**, and **system reliability**. Focused on **delivering high-throughput, secure, and maintainable backend systems**.

Experience

Infosys Limited | Software Engineer - Backend & AI

July 2025-Present

- **Engineered robust FastAPI microservices** to expose RAG and AI agent capabilities via **REST endpoints**; **optimized** request/response lifecycles and middleware for high-concurrency **enterprise-grade AI traffic (RAG)**
- **Developed** reusable tool APIs for AI agents, integrating with enterprise systems like **SharePoint and ERP**, **leveraging** Async Python patterns for efficient background task processing and orchestration.
- **Designed** and optimized PostgreSQL and MongoDB schemas for AI state persistence; **implemented complex indexing strategies** to ensure performant data retrieval and transactional integrity at scale.
- **Integrated** OpenAI/Anthropic APIs with streaming support; managed token limits and **cost monitoring while implementing robust** authentication and authorization middleware for secure AI services.
- **Containerized** services using Docker and managed CI/CD pipelines; **utilized observability tools to track API performance**, debugging, and latency metrics for **production-ready AI deployments**.

Projects

Enterprise AI Integration API | *FastAPI, PostgreSQL, Docker, Async, OpenAI, LangChain*

Jan 2025

- **Problem:** Lack of a centralized API layer for AI agents to query enterprise data securely and efficiently.
- **Built** a RESTful API service to abstract complex RAG logic; integrated **PostgreSQL** for structured **data management** and Qdrant for semantic search **with metadata filtering for secure, role-based access**. control across different user departments.
- **Implemented** asynchronous background workers for high-volume requests; **optimized LLM streaming endpoints** to handle concurrent user inputs **while enforcing robust Pydantic schemas for all payloads**.
- **Developed** security middleware for JWT-based AuthN/AuthZ; deployed in **Docker** environments with automated CI/CD **pipelines** for deployment to enterprise servers, ensuring consistent uptime.
- **Achieved <300ms** API response time **by optimizing database indexing and integrating** caching strategies for frequent AI agent tool calls, enhancing overall system throughput.

Skills

Backend Engineering: Python, FastAPI, REST API Design, AsyncIO, Middleware, OAuth, Authz, Microservices

Databases: PostgreSQL, MongoDB, SQL Optimization, Indexing, Transactions, Vector DBs (Qdrant, Chroma)

AI & Integration: LLM API Consumption (OpenAI/Anthropic), RAG, Streaming, Prompt Engineering, Guardrails

DevOps & Tools: Docker, Git, GitHub Actions CI/CD, Enterprise Integration, SharePoint, ERP, API Tooling

Performance: Background Tasks, Queuing (Celery/Redis), High-Throughput API Serving, Latency Tuning

Core Skills: Production-quality Python, Complex SQL, Software Architecture, Enterprise System Connectivity

Achievements

- **Leetcode (Max Rating 1797)** | Top 5% in India Among 100,000 active users.  [Leetcode](#)
- **3 Star - Codechef (Max Rating 1628)** | Top 5% Among 170K+ Active Users  [Codechef](#)
- **Pupil - Codeforces (Max Rating 1206)** | Top 10% Among 70K+ Active Users  [Codeforces](#)
- **1st Place in VNPS** | National Level Project Showcase
- **3rd place in CodeCraze** | Data Structures and Algorithm Implementation Based Competition

Education

Vidyavardhini's College of Engineering And Technology

May 2025

Bachelor of Engineering in Computer Science - **CGPA - 8.5**